

Noise Pollution and Human Capital Formation

Literature Overview

Noise pollution is a omnipresent environmental externality, yet its consequences for human capital formation remain largely unexplored relative to other pollutants. We situate our contribution within three literatures: causal evidence on other environmental pollutants and human capital, the economics of noise as an externality affecting well-being, productivity, and health, and the small existing body of work on noise and human capital specifically, whose identification challenges we aim to address.

Other Environmental Pollutants and Human Capital

A parallel economics literature documents comparable causal effects of other pollutants — chiefly particulate matter and extreme heat — on human capital formation. A recent meta-analysis pools the causal evidence across 28 studies and finds an average penalty of -0.022 SD (Nguyen 2026). Effects operate across the life course: early-life particulate exposure lowers long-run schooling (Sanders 2012) and earnings (Isen et al. 2017), while temporary exposure during schooling or high-stakes exams lowers test scores, with downstream effects on educational attainment and wages (Ebenstein et al. 2016). Similar effects are documented for extreme heat (Park et al. 2020). For the pollution channel, a smaller but growing literature shows the damage is at least partly abatable through filtration (Gilraine 2025; Bonan et al. 2024). This literature establishes that environmental disturbances can leave lasting marks on human capital. The question we ask is whether the same holds for noise.

Noise, Wellbeing, and Productivity

The literature on the effects of noise pollution is considerably smaller. Economic scholars have studied it as an environmental externality affecting well-being, productivity, and asset prices. Van Praag and Baarsma (2005) use happiness surveys to value aircraft noise around Amsterdam Airport, showing that noise reduces subjective well-being and that these welfare losses are not necessarily fully capitalized into house prices. Dean (2024) provides experimental

evidence from textile workers in Kenya that a 7 dB increase in workplace noise reduces output by about 3 percent, impairs cognitive function, and that workers do not fully recognize the productivity cost of noisy working conditions. Hener (2022) exploits daily variation in aircraft landing approaches at Frankfurt Airport and finds that higher aircraft noise increases violent crime, particularly physical assaults. Krekel and Zerrahn (2017) combine SOEP household panel data with wind turbine locations in Germany and find that nearby turbine construction reduces residential life satisfaction — with effects concentrated within about 4 km and fading after at most five years. Two studies isolate the effect of noise on property values specifically, with consistent implications in opposite directions: Allroggen et al. (2025) show that technical flight path adaptation at three US airports decreased property values where it increased noise, while Moretti and Wheeler (2025) find that the construction of highway noise barriers in Florida — which reduce noise — increased affected property values.

Noise, Health, and Sleep

Research in public health and epidemiology has found impacts on metabolic, cardiovascular, and mental health (Welch et al. 2023). Most relevant to our research is the evidence on sleep: a systematic review of environmental noise and sleep finds transportation noise disturbs objectively measured sleep, and disturbs self-reported sleep specifically when survey questions reference noise explicitly (Basner and McGuire 2018). In a direct demonstration of the objective effect, Basner (2008) uses a laboratory design and shows that nocturnal aircraft noise significantly increases objectively measured next-morning sleepiness. Griefahn et al. (2006) compare road, rail, and aircraft noise in a sleep-laboratory setting and find that transport noise worsens subjective sleep quality and reaction time, while physiological sleep disruption differs by noise source. Bodin et al. (2015) study combined road and railway traffic noise in Malmö and find that access to a quiet side of the dwelling is associated with lower noise annoyance, fewer concentration problems, and better sleep when the bedroom faces the quiet side. Trimmel et al. (2012) experimentally expose participants to low-intensity aircraft or neighbourhood noise and find impaired text learning together with physiological stress responses, especially under aircraft noise.

A set of studies exploits quasi-random variation to strengthen causal claims on health (Nyberg 2025). Boes et al. (2013) exploit unexpected changes in flight regulations at Zurich airport and find that higher aircraft noise increases self-reported sleeping problems and headaches. Beghelli et al. (2023) use unannounced changes in early-morning aircraft landings at Heathrow and find reductions in medication expenditures for central nervous system and respiratory conditions when noise exposure falls. Argys et al. (2020) utilize exogenous variation in a US flight path policy concentrating air traffic to show that the resulting noise increased low birth weight occurrences. Wang et al. (2022) study aircraft noise in New York and find increases in insomnia, adult cardiovascular disease, and depression, as well as mental-health- and substance-use-related healthcare utilization. Krekel et al. (2023) find no corresponding evidence that turbines worsen general, mental, or physical health, doctor visits, or sleep. Greenhill (2026) utilizes changes

in wind direction redirecting train noise exposure in California to identify strong impacts on infant health. Lindgren (2021) evaluates a Swedish road traffic noise mitigation program and finds delayed but substantial reductions in cardiovascular diagnoses, primarily driven by hypertension. Zou (2020) studies US wind farm installations and finds increases in suicide rates, with evidence suggesting sleep disruption from low-frequency noise as a possible mechanism.

Noise and Human Capital

There also exists a small body of work on the impacts of noise on human capital formation. A key challenge in identifying such effects is causality. In particular, we expect noise pollution to be priced in the rental market, causing residential sorting. In equilibrium, individuals with higher tolerance or lower income will be exposed to noise at a higher rate. If this is not accounted for, estimates of the impact of noise pollution will not be representative of the impact on the average citizen or be confounded with other relevant behavior. Avoidance behavior might also be problematic if it is heterogeneous across third factors with a direct link to health. Additionally, exposure to noise might be correlated with air pollution and (extreme) weather, causing further confounding. The existing body of research outlined below does not fully address these challenges. Haines et al. (2001) study schoolchildren near London Heathrow and find that chronic aircraft noise exposure is associated with higher annoyance and poorer reading comprehension, but not with worse mental health or elevated cortisol. Hygge et al. (2002) exploit the opening and closure of Munich airports and find that aircraft noise impairs children’s reading and long-term memory, while cognitive performance improves after noise exposure ceases. Clark et al. (2010) use the cross-country RANCH data and find that aircraft noise is linearly associated with worse recognition memory in children, while road traffic noise shows different and more mixed associations with episodic memory. Klatte et al. (2017) study primary school children around Frankfurt Airport and find that even comparatively moderate aircraft noise exposure is associated with lower reading performance, higher annoyance, and worse quality of life.

None of these four studies resolves all challenges simultaneously. Haines et al. (2001) match high- and low-noise schools on age, existing school-level sound insulation, deprivation, and ethnicity, but the matching operates at the school level — it does not address self-selection of families into noisy versus quiet homes within catchment areas — and the study includes no measure of avoidance behavior (e.g., insulation or quiet-facing bedrooms) or of co-varying air pollution and weather. Hygge et al. (2002) come closest to addressing sorting: because the source of noise variation is the relocation of Munich’s airport rather than families relocating, exposure changes independently of residential choice. But the design still cannot separate noise from co-occurring changes in aircraft-related air pollution at the same sites, includes no avoidance-behavior measure, and — as Klatte et al. (2017) note in their own critique of this study — relies on small, unevenly attriting subsamples and an undocumented exclusion of poor readers from the reading-score analysis. Clark et al. (2010) improve on individual-level confounding by adjusting for mother’s education, employment, crowding, and home ownership,

and are unusual in directly controlling for classroom window glazing as a proxy for sound insulation — but the design is cross-sectional, so sorting into noise-exposed catchment areas on unobserved family traits is not resolved by covariate adjustment, and air pollution and weather remain uncontrolled; Klatte et al. (2017) further note that they do not account for clustering of children within schools and classes, risking overstated significance. Klatte et al. (2017) address that clustering problem with multilevel models and combine individual- and class-level socioeconomic controls more completely than prior work, and separately control for acute noise at the time of testing to avoid conflating chronic and acute exposure — but their design is likewise cross-sectional across noise-exposed areas, so residential sorting is addressed only insofar as it correlates with the included socioeconomic covariates, and neither avoidance behavior nor air pollution and weather are measured. We address these remaining sorting, avoidance-behavior, and confounding concerns by exploiting the staggered, plausibly exogenous timing of the BoViT insulation and noise-barrier rollout described below.

Data

Exposure

We identify exposure to noise pollution using the staggered rollout of mitigation technology. The Swedish transportation authority (Trafikverket) provides dwellings and neighborhoods with sound insulation and noise barriers, respectively. Obtaining data on the provision of these measures is a prerequisite for our analysis. The BoViT noise mitigation program includes information on all provided insulation and barriers, most notably identifying the exact dwellings that received insulation, as well as the date of provision. This data is not publicly available and will have to be requested from Trafikverket. A smaller subset covering only noise barriers is publicly available. Inspection of this data revealed that construction dates are missing for a large share of barriers; these gaps could potentially be closed with Trafikverket's assistance, but the BoViT data should be prioritized where possible. Supplementary data on dwelling, road, and rail positions is publicly available. We can also request cross-sectional modeled noise exposure for the entire country, as well as historical traffic density data at measurement points. This will allow us to control for background noise exposure and traffic density in our analysis.

Data overview

Legend: *Public* = publicly available now | *Request* = requires a data request to Trafikverket
 | *Partial* = some records public, remainder on request

Table 1: Data sources for exposure, auxiliary, and control variables. Outcome data (individual/school-level human capital measures) is not yet listed and needs to be added.

Role in analysis	Dataset	Source	Status	Access
Exposure	BoViT project data (incl. dwelling-level noise insulation) ¹	Trafikverket	Request	Request from Trafikverket
Exposure	Sound barriers — roads	Trafikverket	Public	Lastkajen entry 630
Exposure	Sound barriers — railway	Trafikverket	Public	Lastkajen entry 991
Auxiliary	Dwelling positions	Landmateriet	Public	Landmateriet geotorget
Auxiliary	INSPIRE street network plan	Trafikverket	Public	Lastkajen
Auxiliary	INSPIRE railway network plan	Trafikverket	Public	Lastkajen
Auxiliary	BoViT noise simulation data	Trafikverket	Request	Request from Trafikverket
Control	Traffic density (årsmedeldygnstrafik)	Trafikverket	Partial	Lastkajen (partial); remainder on request

¹Previously used by Lindgren (2021).

References

- Allroggen, Florian, R. John Hansman, Christopher Knittel, Jing Li, Xibo Wan, and Juju Wang. 2025. *Planes Overhead: How Airplane Noise Impacts Home Values*. No. w34431. National Bureau of Economic Research. <https://doi.org/10.3386/w34431>.
- Argys, Laura M., Susan L. Averett, and Muzhe Yang. 2020. “Residential Noise Exposure and Health: Evidence from Aviation Noise and Birth Outcomes.” *Journal of Environmental Economics and Management* 103 (September): 102343. <https://doi.org/10.1016/j.jeem.2020.102343>.
- Basner, M. 2008. “Nocturnal Aircraft Noise Exposure Increases Objectively Assessed Daytime Sleepiness.” *Somnologie - Schlafforschung Und Schlafmedizin* 12 (2): 110–17. <https://doi.org/10.1007/s11818-008-0338-8>.
- Basner, Mathias, and Sarah McGuire. 2018. “WHO Environmental Noise Guidelines for the European Region: A Systematic Review on Environmental Noise and Effects on Sleep.” *International Journal of Environmental Research and Public Health* 15 (3): 519. <https://doi.org/10.3390/ijerph15030519>.
- Beghelli, Silvia, Augustin De Coulon, and Mary O’Mahony. 2023. “Health Benefits of Reducing Aircraft Pollution: Evidence from Changes in Flight Paths.” *Journal of Population Economics* 36 (4): 2581–607. <https://doi.org/10.1007/s00148-023-00964-y>.
- Bodin, Theo, Jonas Björk, Jonas Ardö, and Maria Albin. 2015. “Annoyance, Sleep and Concentration Problems Due to Combined Traffic Noise and the Benefit of Quiet Side.” *International Journal of Environmental Research and Public Health* 12 (2): 1612–28. <https://doi.org/10.3390/ijerph120201612>.
- Boes, Stefan, Stephan Nüesch, and Steven Stillman. 2013. “AIRCRAFT NOISE, HEALTH, AND RESIDENTIAL SORTING: EVIDENCE FROM TWO QUASI-EXPERIMENTS.” *Health Economics* 22 (9): 1037–51. <https://doi.org/10.1002/hec.2948>.
- Bonan, Jacopo, Francesco Granella, Stefania Renna, and Luis Sarmiento. 2024. *The Effect of Air Purifiers in Schools*. RFF-CMCC European Institute on Economics and the Environment (EIEE).
- Clark, Charlotte, Rocio Martin, Elise Van Kempen, et al. 2010. “The Effects of Road Traffic and Aircraft Noise Exposure on Children’s Episodic Memory: The RANCH Project.” *Noise and Health* 12 (49): 244. <https://doi.org/10.4103/1463-1741.70503>.
- Dean, Joshua T. 2024. “Noise, Cognitive Function, and Worker Productivity.” *American*

- Economic Journal: Applied Economics* 16 (4): 322–60. <https://doi.org/10.1257/app.20220532>.
- Ebenstein, Avraham, Victor Lavy, and Sefi Roth. 2016. “The Long-Run Economic Consequences of High-Stakes Examinations: Evidence from Transitory Variation in Pollution.” *American Economic Journal: Applied Economics* 8 (4): 36–65. <https://doi.org/10.1257/app.20150213>.
- Gilraine, Michael. 2025. “Air Filters, Pollution, and Student Achievement.” *Journal of Human Resources* 60 (5): 1469–506. <https://doi.org/10.3368/jhr.0421-11642R2>.
- Greenhill, Simon. 2026. *Noise Pollution and Infant Health*.
- Griefahn, Barbara, Anke Marks, and Sibylle Robens. 2006. “Noise Emitted from Road, Rail and Air Traffic and Their Effects on Sleep.” *Journal of Sound and Vibration* 295 (1-2): 129–40. <https://doi.org/10.1016/j.jsv.2005.12.052>.
- Haines, M. M., S. A. Stansfeld, R. F. S. Job, B. Berglund, and J. Head. 2001. “Chronic Aircraft Noise Exposure, Stress Responses, Mental Health and Cognitive Performance in School Children.” *Psychological Medicine* 31 (2): 265–77. <https://doi.org/10.1017/S0033291701003282>.
- Hener, Timo. 2022. “Noise Pollution and Violent Crime.” *Journal of Public Economics* 215 (November): 104748. <https://doi.org/10.1016/j.jpubeco.2022.104748>.
- Hygge, Staffan, Gary W. Evans, and Monika Bullinger. 2002. “A Prospective Study of Some Effects of Aircraft Noise on Cognitive Performance in Schoolchildren.” *Psychological Science* 13 (5): 469–74. <https://doi.org/10.1111/1467-9280.00483>.
- Isen, Adam, Maya Rossin-Slater, and W. Reed Walker. 2017. “Every Breath You Take—Every Dollar You’ll Make: The Long-Term Consequences of the Clean Air Act of 1970.” *Journal of Political Economy* 125 (3): 848–902. <https://doi.org/10.1086/691465>.
- Klatte, Maria, Jan Spilski, Jochen Mayerl, Ulrich Möhler, Thomas Lachmann, and Kirstin Bergström. 2017. “Effects of Aircraft Noise on Reading and Quality of Life in Primary School Children in Germany: Results From the NORAH Study.” *Environment and Behavior* 49 (4): 390–424. <https://doi.org/10.1177/0013916516642580>.
- Krekel, Christian, Johannes Rode, and Alexander Roth. 2023. *Do Wind Turbines Have Adverse Health Impacts?* {{IZA Discussion Papers}} No. 16505. Institute of Labor Economics (IZA).
- Krekel, Christian, and Alexander Zerrahn. 2017. “Does the Presence of Wind Turbines Have Negative Externalities for People in Their Surroundings? Evidence from Well-Being

- Data.” *Journal of Environmental Economics and Management* 82 (March): 221–38. <https://doi.org/10.1016/j.jeem.2016.11.009>.
- Lindgren, Samuel. 2021. *A Sound Environment: Health Effects of Traffic Noise Mitigation*. Working {{Papers}} 2021:10. Swedish National Road & Transport Research Institute (VTI). <https://doi.org/None>.
- Moretti, Enrico, and Harrison Wheeler. 2025. *The Traffic Noise Externality: Costs, Incidence and Policy Implications*. No. w34298. National Bureau of Economic Research. <https://doi.org/10.3386/w34298>.
- Nguyen, Tuan D. 2026. *The Effect of Air Pollution on Student Achievement: A Systematic Review and Meta-Analysis of the Causal Evidence*. Edworkingpapers.com. <https://doi.org/10.26300/QEKG-T772>.
- Nyberg, Erik. 2025. *Air and Noise Pollution: A State-of-the-Art Review of Causal Health Outcomes*. 2025:6.
- Park, R. Jisung, A. Patrick Behrer, and Joshua Goodman. 2020. “Learning Is Inhibited by Heat Exposure, Both Internationally and Within the United States.” *Nature Human Behaviour* 5 (1): 19–27. <https://doi.org/10.1038/s41562-020-00959-9>.
- Sanders, Nicholas J. 2012. “What Doesn’t Kill You Makes You Weaker: Prenatal Pollution Exposure and Educational Outcomes.” *Journal of Human Resources* 47 (3): 826–50. <https://doi.org/10.1353/jhr.2012.0018>.
- Trimmel, Michael, Jürgen Atzlsdorfer, Nina Tupy, and Karin Trimmel. 2012. “Effects of Low Intensity Noise from Aircraft or from Neighbourhood on Cognitive Learning and Electrophysiological Stress Responses.” *International Journal of Hygiene and Environmental Health* 215 (6): 547–54. <https://doi.org/10.1016/j.ijheh.2011.12.007>.
- Van Praag, Bernard M. S., and Barbara E. Baarsma. 2005. “Using Happiness Surveys to Value Intangibles: The Case of Airport Noise.” *The Economic Journal* 115 (500): 224–46. <https://doi.org/10.1111/j.1468-0297.2004.00967.x>.
- Wang, Scarlett Sijia, Sherry Glied, Sharifa Williams, Brian Will, and Peter Alexander Muennig. 2022. “Impact of Aeroplane Noise on Mental and Physical Health: A Quasi-Experimental Analysis.” *BMJ Open* 12 (5): e057209. <https://doi.org/10.1136/bmjopen-2021-057209>.
- Welch, David, Daniel Shepherd, Kim N. Dirks, and Ravi Reddy. 2023. “Health Effects of Transport Noise.” *Transport Reviews* 43 (6): 1190–210. <https://doi.org/10.1080/01441647.2023.2206168>.

Zou, Eric. 2020. *Wind Turbine Syndrome: The Impact of Wind Farms on Suicide*.